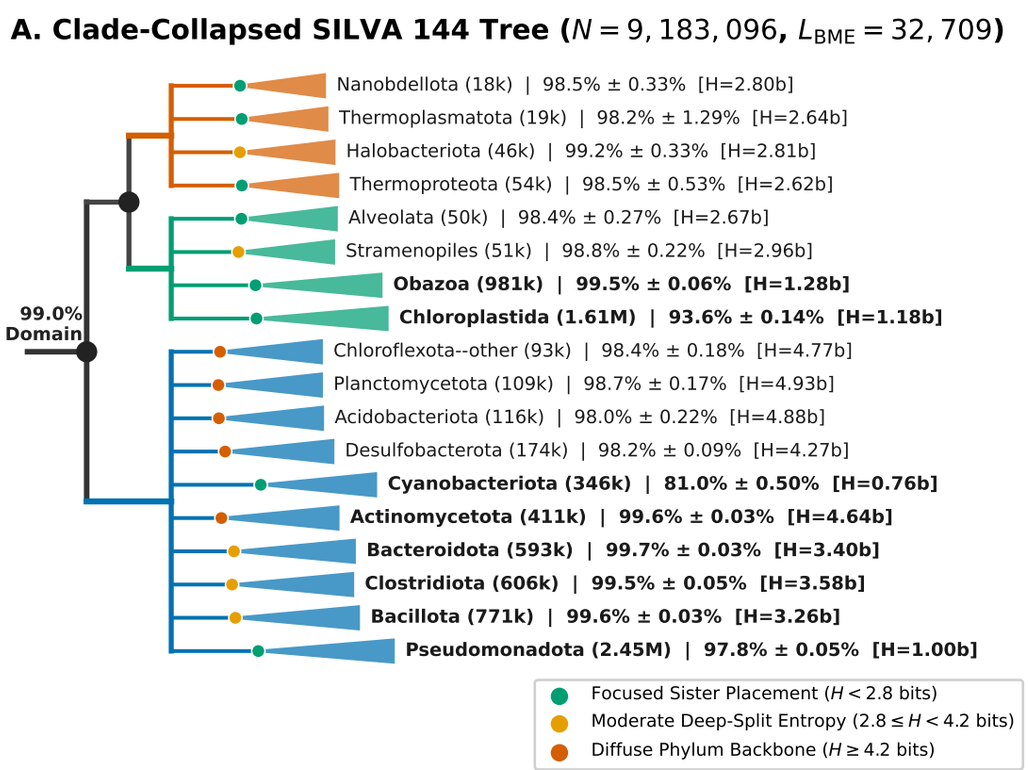
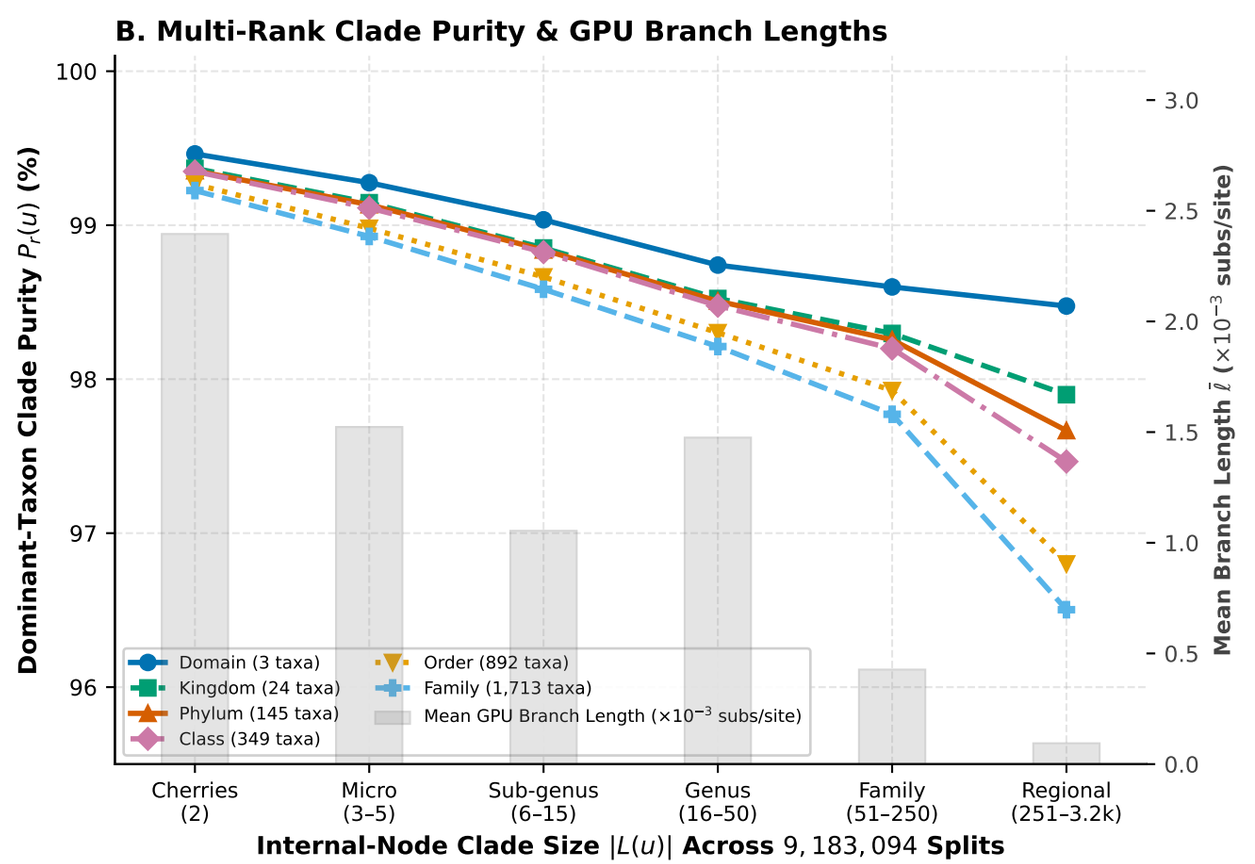
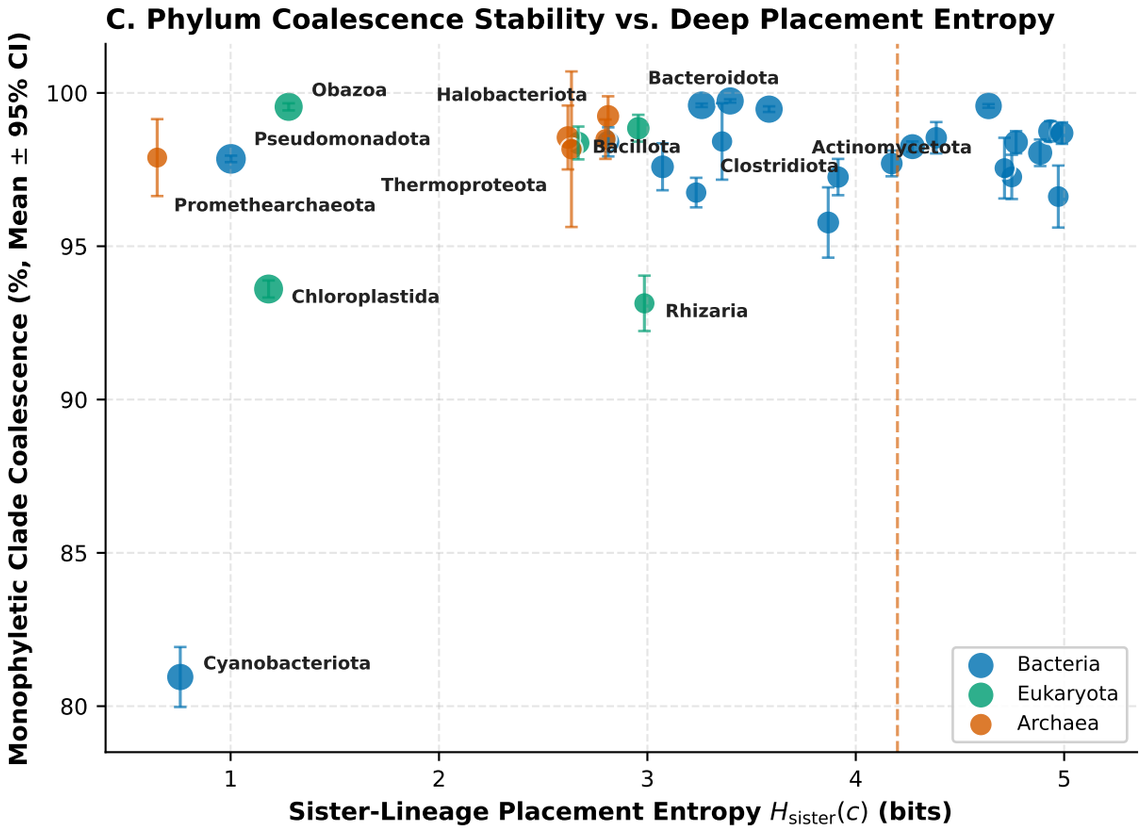
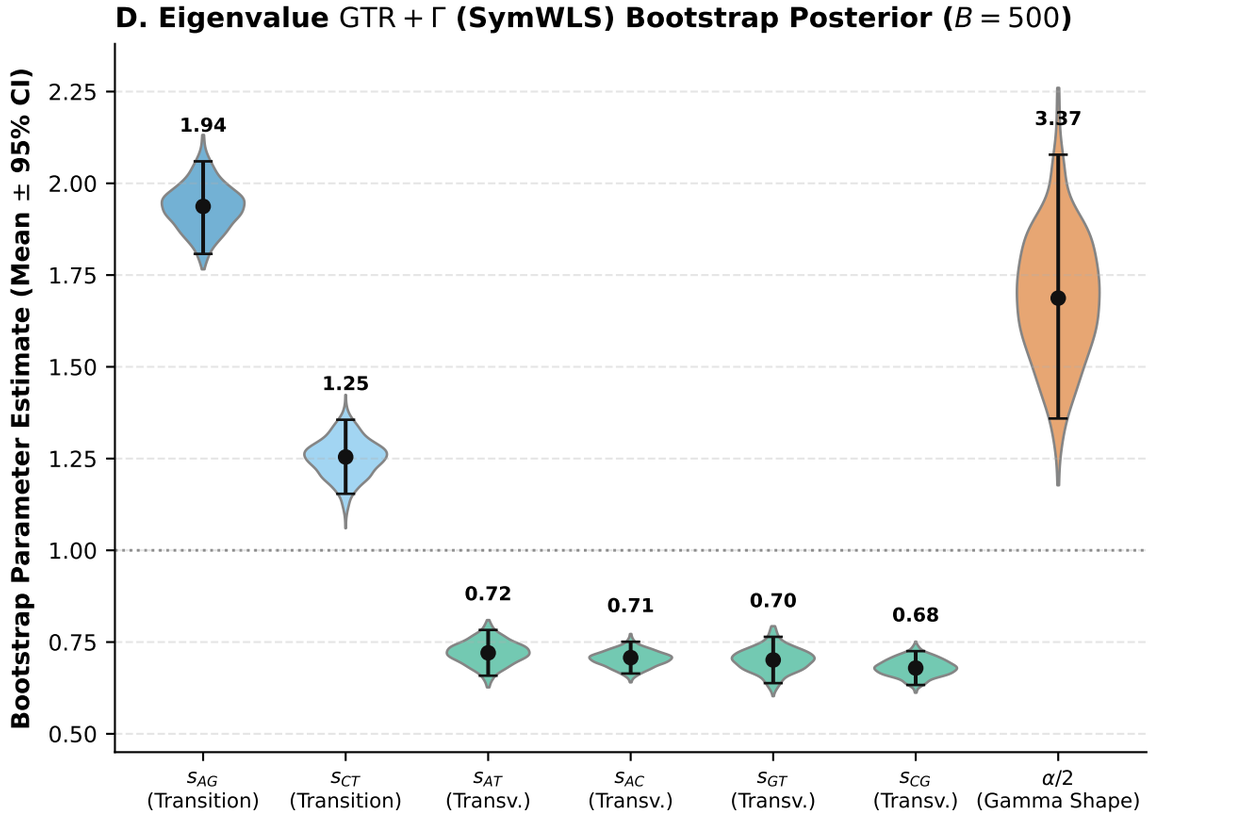

A. Clade-Collapsed SILVA 144 Tree ($N = 9,183,096$, $L_{\mathrm{BME}} = 32,709$)
99.0%
Domain
Nanobdellota (18k) | 98.5% ± 0.33% [H=2.80b]
Thermoplasmatota (19k) | 98.2% ± 1.29% [H=2.64b]
Halobacteriota (46k) | 99.2% ± 0.33% [H=2.81b]
Thermoproteota (54k) | 98.5% ± 0.53% [H=2.62b]
Alveolata (50k) | 98.4% ± 0.27% [H=2.67b]
Stramenopiles (51k) | 98.8% ± 0.22% [H=2.96b]
Obazoa (981k) | 99.5% ± 0.06% [H=1.28b]
Chloroplastida (1.61M) | 93.6% ± 0.14% [H=1.18b]
Chloroflexota--other (93k) | 98.4% ± 0.18% [H=4.77b]
Planctomycetota (109k) | 98.7% ± 0.17% [H=4.93b]
Acidobacteriota (116k) | 98.0% ± 0.22% [H=4.88b]
Desulfobacterota (174k) | 98.2% ± 0.09% [H=4.27b]
Cyanobacteriota (346k) | 81.0% ± 0.50% [H=0.76b]
Actinomycetota (411k) | 99.6% ± 0.03% [H=4.64b]
Bacteroidota (593k) | 99.7% ± 0.03% [H=3.40b]
Clostridiota (606k) | 99.5% ± 0.05% [H=3.58b]
Bacillota (771k) | 99.6% ± 0.03% [H=3.26b]
Pseudomonadota (2.45M) | 97.8% ± 0.05% [H=1.00b]
Focused Sister Placement ($H < 2.8$ bits)
Moderate Deep-Split Entropy ($2.8 \le H < 4.2$ bits)
Diffuse Phylum Backbone ($H \ge 4.2$ bits)


B. Multi-Rank Clade Purity & GPU Branch Lengths
Dominant-Taxon Clade Purity $P_r(u)$ (%)
Mean Branch Length $\bar{\ell}$ ($\times 10^{-3}$ subs/site)
100
99
98
97
96
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Domain (3 taxa)
Kingdom (24 taxa)
Phylum (145 taxa)
Class (349 taxa)
Order (892 taxa)
Family (1,713 taxa)
Mean GPU Branch Length ($\times 10^{-3}$ subs/site)
Cherries (2)
Micro (3–5)
Sub-genus (6–15)
Genus (16–50)
Family (51–250)
Regional (251–3.2k)
Internal-Node Clade Size $|L(u)|$ Across $9,183,094$ Splits


C. Phylum Coalescence Stability vs. Deep Placement Entropy
Monophyletic Clade Coalescence (%, Mean ± 95% CI)
100
95
90
85
80
1
2
3
4
5
Sister-Lineage Placement Entropy $H_{\mathrm{sister}}(c)$ (bits)
Obazoa
Pseudomonadota
Promethearchaeota
Halobacteriota
Bacteroidota
Bacillota
Thermoproteota
Clostridiota
Actinomycetota
Chloroplastida
Rhizaria
Cyanobacteriota
Bacteria
Eukaryota
Archaea


D. Eigenvalue GTR + Γ (SymWLS) Bootstrap Posterior ($B = 500$)
Bootstrap Parameter Estimate (Mean ± 95% CI)
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
1.94
1.25
0.72
0.71
0.70
0.68
3.37
$s_{AG}$ (Transition)
$s_{CT}$ (Transition)
$s_{AT}$ (Transv.)
$s_{AC}$ (Transv.)
$s_{GT}$ (Transv.)
$s_{CG}$ (Transv.)
$\alpha/2$ (Gamma Shape)